SHEENA P

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EDUCATION

Bachelor of Computer Science And Engineering (Artificial Intelligence and Machine Learning), 2024 – 2028
Chennai Institute of Technology
- Cumulative Grade Point Average (CGPA) - 8.73

EXPERIENCE

Research Intern, *NIT Puducherry* 04/2026 – 06/2026
Built an end-to-end medical QA system using oncology datasets, MRL embeddings, LAQA, and medical LLMs (Medgemma). Performed retrieval optimization, answer generation, and ablation studies to assess the contribution of individual pipeline modules.

PROJECTS

CloudSentinel AI – Intelligent Cloud Resource Monitoring & Cost Anomaly Detection
- Developed an ML-based cloud resource anomaly detection system using **Isolation Forest** and resource-specific feature engineering. Deployed containerized inference using **AWS Lambda** and **Amazon ECR**, integrated **Amazon S3** for input/output storage, configured **Amazon CloudWatch** for execution monitoring, and established **AWS Budgets** alerts for cost tracking.
- Tools: Python, Scikit-learn, Pandas, AWS Lambda, Amazon S3, Amazon ECR, Amazon CloudWatch, AWS Budgets, Docker.

Yojana Mitra – AI Powered Government Scheme Navigator (RAG + LLM)
- Developed an AI-powered platform that helps citizens discover eligible government welfare schemes using **Retrieval-Augmented Generation (RAG)**, combining semantic search with **Large Language Models** to provide personalized recommendations, eligibility explanations, scheme comparison, application guidance, and document requirement generation with citation-based responses.
- Tools: Python, FastAPI, React, HuggingFace Sentence Transformers, Claude LLM, LangChain.

Al Based Breast Cancer Detection
- Developed an AI-based breast cancer detection system using a custom CNN with attention mechanisms to classify mammographic images into Normal, Benign, or Malignant. Achieved high test accuracy with <5-second inference time and built a user-friendly GUI that displays class predictions with confidence scores.
- Tools: Python, Tensorflow, OpenCV, Numpy, Pandas,Matplotlib,Seaborn, tkinter

TourPulse — MLOps-Driven Tourism Demand Forecasting Platform
- Built a full-stack MLOps platform forecasting tourism demand across 15 Indian cities using ensemble ML models, live weather integration, SHAP explainability, and automated drift detection with a React dashboard and FastAPI backend deployed on cloud.
- Tools: Python, Random Forest, SHAP, FastAPI, React, Render, Vercel

SKILLS

Programming Languages
C++, SQL

Web Development
HTML, CSS, React.js

Machine Learning & Data Science
Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn

Tools & Platforms
Git, AWS, Linux, Docker

CERTIFICATIONS

- Machine Learning - Internshala Trainings
- CLF-CO2 AWS Certified Cloud Practitioner
- Claude with Anthropic API Training

ACHIEVEMENTS

Solved 600+ Problems in Leetcode with 1700+ rating.

National Finalist: Neurohack'26, *IIT Madras*

Published 2 IEEE Conference Research Papers in international conferences (NELEX 2026 & ICICV 2026).

Research

Network- and Cost-Aware Adaptive Cloud–Edge Computer Vision Framework Using Machine Learning for Latency-Critical Applications

Interpretable Multi-Organ Dynamics Prediction via Graph Attention Networks and Counterfactual Simulation.

Volunteering

Research and Innovation Lead, ATHERA
Mentored members in research and emerging technologies while coordinating innovation-focused projects and technical events.

Cloud Computing Workshop Speaker, MAGNUS Symposium
Delivered a cloud computing workshop to students from multiple colleges as part of the AIML Department Symposium, introducing core cloud concepts and practical applications.